

## Suppressed background — labeling clinical EEG reports

We labeled every report for whether it explicitly calls the EEG background **suppressed** — i.e. generalized or diffuse suppression of the background activity. Three-way call: the report says so (**present**), the report says suppression is absent (**explicitly absent**), or it isn't mentioned (**not mentioned**). Vasily's suggested prompt, used verbatim, grammar-constrained so the answer is exactly one of those three. MedGemma-27B (Q4\_K\_S), temperature 0.

The prompt deliberately keeps this separate from burst-suppression: a report that describes burst-suppression, suppression-burst, burst-attenuation, or brief event-related suppression is **not** counted as a suppressed background unless it also explicitly says the overall background is suppressed.

It ran on Fraser Health (45,545 reports) and three Harvard hospitals — MGH, BWH, BCH (~100,600; BIDMC has no report text). About 1.3% of Harvard reports are longer than the context window and returned nothing; they are excluded.

### Is there a matching Harvard variable?

No. Harvard's bs column is *burst*-suppression, a different pattern; low voltage is amplitude, also different. There is no column for a plainly suppressed background, so this field is our labels only, with no head-to-head against Harvard. How Harvard codes the columns it does have is in the companion note reports/harvard\_llm\_labeling.md.

### What we found

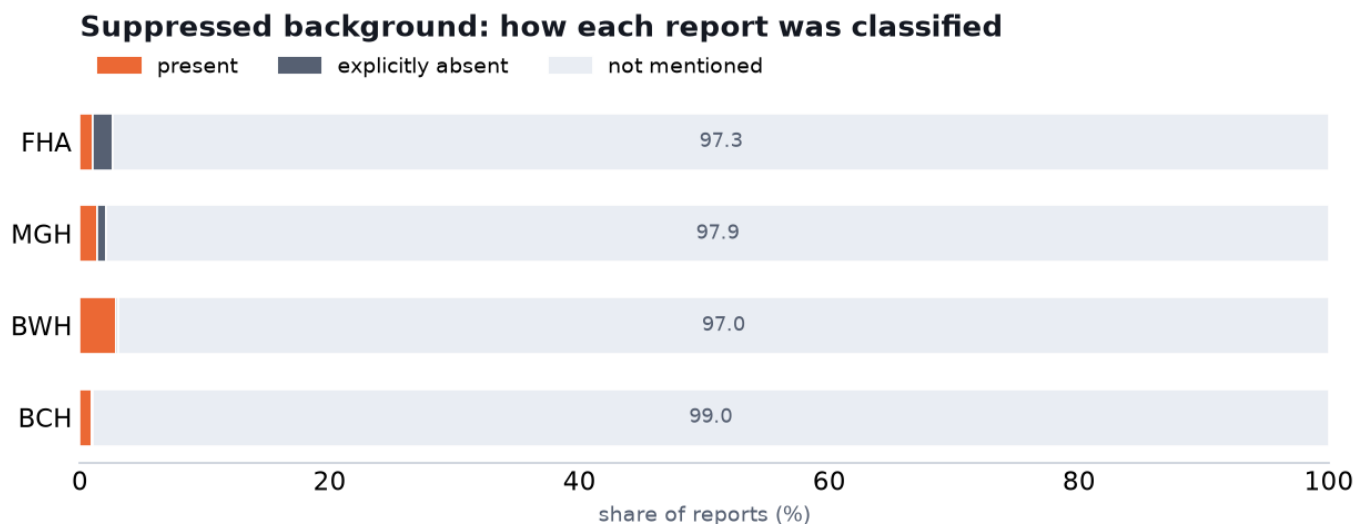


Figure 1: Suppressed background — distribution across datasets

| dataset       | present | explicitly absent | not mentioned | reports |
|---------------|---------|-------------------|---------------|---------|
| Fraser Health | 1.0%    | 1.7%              | 97.4%         | 45,545  |
| MGH           | 1.4%    | 0.7%              | 97.9%         | 54,823  |
| BWH           | 2.9%    | 0.1%              | 97.0%         | 20,691  |
| BCH           | 0.9%    | 0.1%              | 99.0%         | 25,022  |

Suppressed background behaves like an ordinary rare finding, and it behaves the same way across both cohorts: present in roughly 1–3% of reports, almost never explicitly called absent, and otherwise not mentioned. Unlike discontinuous background (where the Harvard reports comment on continuity far more than Fraser Health's do), here there's no cohort quirk — the reporting style is consistent, and the present rates line up. BWH is slightly higher (2.9%), in line with its generally higher critical-care rates —

it also tops the triphasic and burst-suppression counts.

## Caveats

- No Harvard column to check against, so this is our own labeling with no external reference.
- Kept strictly separate from burst-suppression per the prompt — a report is only counted here if it calls the *overall background* suppressed, not because suppression occurs inside another pattern.
- About 1.3% of Harvard reports were too long for the context window and returned no label; they are excluded.

## Reproduce

```
# labeling (on the cluster)
bash slurm/submit_background.sh           # Fraser Health, all background fields
bash slurm/submit_background_harvard.sh   # MGH / BWH / BCH
# distribution + chart
python -m analysis.background_dist        # results/background/dist_summary.json
python -m analysis.background_field_charts # figures/dist_suppressed.png
```

Labeler and prompt: src/cpu/background.py (field suppressed). Companion note on the Harvard fields: reports/harvard\_llm\_labeling.md.